```
import pandas as pd

df = pd.read_csv("/content/CO2 Emissions_Canada.csv")
df.head()
```

	Make	Model	Vehicle Class	Engine Size(L)	Cylinders	Transmission	Fuel Type	Fuel Consumption City (L/100 km)	Fuel Consumption Hwy (L/100 km)	Fuel Consumption Comb (L/100 km)	Fuel Consumption Comb (mpg)	CO2 Emissions(g/km)
0	ACURA	ILX	COMPACT	2.0	4	AS5	Z	9.9	6.7	8.5	33	196
1	ACURA	ILX	COMPACT	2.4	4	M6	Z	11.2	7.7	9.6	29	221
2	ACURA	ILX HYBRID	COMPACT	1.5	4	AV7	Z	6.0	5.8	5.9	48	136
3	ACURA	MDX 4WD	SUV - SMALL	3.5	6	AS6	Z	12.7	9.1	11.1	25	255
4	ACURA	RDX AWD	SUV - SMALL	3.5	6	AS6	Z	12.1	8.7	10.6	27	244

```
X = df[["Engine Size(L)", "Cylinders", "Fuel Consumption Comb (L/100 km)"]]
y = df["CO2 Emissions(g/km)"]
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
from sklearn.linear_model import LinearRegression

lr = LinearRegression()
lr.fit(X_train_scaled, y_train)

y_pred_lr = lr.predict(X_test_scaled)
```

```
from sklearn.linear_model import Lasso

lasso = Lasso(alpha=0.1)
lasso.fit(X_train_scaled, y_train)

y_pred_lasso = lasso.predict(X_test_scaled)
```

```
from sklearn.linear_model import Ridge
from sklearn.model_selection import GridSearchCV

param_grid = {"alpha": [0.01, 0.1, 1, 10, 100]}

ridge = Ridge()

grid = GridSearchCV(ridge, param_grid, cv=5)
grid.fit(X_train_scaled, y_train)

best_ridge = grid.best_estimator_
best_alpha = grid.best_params_["alpha"]

y_pred_ridge = best_ridge.predict(X_test_scaled)

print("Best Alpha:", best_alpha)
```

```
Best Alpha: 10
```

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

def evaluate(name, y_test, y_pred):
    mae = mean_absolute_error(y_test, y_pred)
    rmse = np.sqrt(mean_squared_error(y_test, y_pred))
    r2 = r2_score(y_test, y_pred)

    print(f"----- {name} -----")
    print("MAE :", mae)
    print("RMSE:", rmse)
    print("R2  :", r2)
    print()

    return mae, rmse, r2
```

```
results = {}

results["Linear Regression"] = evaluate("Linear Regression", y_test, y_pred_lr)
results["Lasso Regression"] = evaluate("Lasso Regression", y_test, y_pred_lasso)
results["Best Ridge Regression"] = evaluate("Best Ridge Regression", y_test, y_pred_ridge)

results_df = pd.DataFrame(results, index=["MAE", "RMSE", "R2 Score"]).T
results_df
```

```
----- Linear Regression -----
MAE : 13.517321294682649
RMSE: 20.540748085335153
R2  : 0.8773348735033225

----- Lasso Regression -----
MAE : 13.534646737575086
RMSE: 20.54241157861251
R2  : 0.8773150046187961

----- Best Ridge Regression -----
MAE : 13.543667005117833
RMSE: 20.541396384652405
R2  : 0.8773271303605337
```

	MAE	RMSE	R2 Score
Linear Regression	13.517321	20.540748	0.877335
Lasso Regression	13.534647	20.542412	0.877315
Best Ridge Regression	13.543667	20.541396	0.877327